RAJ BHARAT RASAL

Senior Multi-Agent AI & Safety Systems Engineer

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PROFESSIONAL SUMMARY

AI Systems Architect and Founder of SecureVerse specializing in multi-agent orchestration, parallel persona fan-out, LLM intent routing, progressive RAG, and cloud-native serverless inference pipelines.

TECHNICAL & ARCHITECTURAL SKILLS

Languages: Python, C++, Rust, Java, TypeScript / JavaScript, SQL (PostgreSQL, SQLite), Bash / Shell Scripting

Distributed Systems & Architecture: Distributed Systems Architecture, Tensor Sharding & Compute Networks, Decoupled Orchestration, Zero-Trust Communication Protocols, Proof-of-Compute Verification, System Design & Scalability, Soroban Smart Contracts (Stellar), Futuristic Systems Design

AI & Machine Learning: Multi-Agent AI Pipelines, AI Safety & Alignment, LLM Intent Classification & Routing, Prompt Engineering & Chains, Progressive RAG & Memory Layers, Adversarial Critic Systems, Ethical AI Systems, Predictive Computational Modeling

FEATURED SYSTEMS & PROJECTS

F.R.I.D.A.Y — Multi-Agent Communication Backend — Lead Backend Architect & Developer

Jun 2026 - Present

Tech Stack: Python, FastAPI, Baileys API (WhatsApp), APScheduler, asyncio, LLM Intent Routing

- Originated at SummerHacks 2026 in a team of 4; currently driving continuous backend evolution and multi-agent infrastructure.
- Designed a strictly decoupled 3-agent architecture (Herald, Nexus, Dispatch) enforcing complete separation of concerns so agents never call each other directly, eliminating circular deadlocks.
- Implemented LLM intent classification with confidence-based routing logic to catch ambiguous queries and trigger clarification flows before dispatching execution tasks.
- Engineered resilient background job scheduling and automated task pipelines with robust fault recovery.
- **Key Architecture:** 3-tier mediator topology; zero direct agent coupling; confidence threshold routing (<0.85 triggers conversational disambiguation); asynchronous message queue via FastAPI.

BioShield AI — Computational Biosafety & Health Resilience Engine — Lead Researcher & Architect

Jan 2026 - Present

Tech Stack: Python, PyTorch, Predictive Modeling, Simulation Engines, FastAPI, Docker

- Engineered predictive computational modeling pipelines for simulating pathogen transmission dynamics and evaluating countermeasure strategies in real time.
- Designed distributed simulation workers to process high-throughput biological interaction datasets with deterministic validation hooks.
- Authored biosafety compliance blueprints and zero-leakage data sandboxing mechanisms for sensitive health research data.
- **Key Architecture:** Event-driven asynchronous worker pool; deterministic simulation checkpointing; isolated compute sandboxes with strict egress filtering.

Cognitive Diplomat — Multi-Agent Negotiation Predictor & RAG — Lead AI Systems Architect

Sep 2026 - Oct 2026

Tech Stack: TypeScript, Express, Ollama LLM, Progressive RAG, In-Memory LRU Cache, Vector-Keyword Hybrid Retrieval

- Engineered a high-performance negotiation next-turn prediction engine orchestrating a 3-persona parallel agent fan-out via Promise.all.
- Implemented a progressive RAG layer over tactical negotiation playbooks with keyword-hybrid retrieval and LRU-bounded memory.
- Enforced deterministic synthesis using temperature=0 and fixed seed assertions with strict confidence score clamping.
- **Key Architecture:** Parallel agent fan-out; deterministic LLM synthesis; 500-entry LRU per-request memory cache; full requirement traceability matrix.

Topcoder AI Profile Video Pipeline — Async Cloud-Native Engine — Cloud & AI Pipeline Architect

Aug 2026 - Sep 2026

Tech Stack: Python, AWS Lambda, MoviePy/FFmpeg, Streamlit, Asynchronous Orchestration, Docker, Render

- Architected an event-driven serverless video processing pipeline that automatically converts raw member intro footage into professional branded Topcoder videos.
- Built an AWS Lambda queue consumer for scalable background transcription, caption overlay, and audio normalization.
- Delivered an interactive Streamlit operational dashboard for cost, performance, and compute budget analysis.
- **Key Architecture:** Decoupled queue-driven serverless architecture; dynamic FFmpeg rendering pipeline; budget and resource observability dashboard.

EXPERIENCE & LEADERSHIP

SecureVerse — Founder & Lead Architect *Oct 2025 - Present*

- Founded SecureVerse to engineer decentralized zero-trust identity verification architectures, AI safety tooling, and privacy-preserving computing systems.
- Formulated end-to-end cryptographic trust chains, automated audit mechanisms, and threat mitigation models recognized in national engineering challenges.
- Directed technology roadmap, leading proof-of-concept development and cross-functional prototype verification.

Independent Tutoring Practice — Mathematics & Physics Tutor (Grades 10–12) *Jun 2024 - Present*

- Taught 50+ high school students, translating complex scientific, mathematical, and algorithmic concepts into clear, structured lessons.
- Developed diagnostic assessment frameworks to identify conceptual bottlenecks and tailored individual curricula to improve performance by an average of 22%.
- Honed mastery of first-principles thinking, communication clarity, and pedagogical system design.

SYSTEM DESIGN SPECIFICATIONS & WHITEPAPERS

HAMIA Architecture Specification: Hierarchical Adaptive Multi-Model Intelligence Architecture (Architecture Whitepaper, Oct 2026)

- A cloud-native fusion system that orchestrates a federation of specialized frontier models (12+) through an 8-stage pipeline. Emphasizes evidence-centric fusion over naive voting, using external ground-truth verifiers (Firecracker sandboxes, SymPy) and a cyclic adversarial critic to prevent multi-model ensembles from amplifying shared bias.

TWELVE Product Documentation: AI-Powered Viva & Interview Assessment System (System Design & Product Specification, Oct 2026)

- A secure, kiosk-based assessment pipeline that replaces human examiner bias with a multi-model AI panel. Features a specialized split-evaluator architecture (Curriculum, Application, Coherence) and a robust legal/compliance strategy for deployment in higher education environments.

EDUCATION

ITM Skills University — Bachelor of Technology (B.Tech) *Aug 2025 - May 2029*

Field: Computer Science Engineering | Score: Undergraduate (In Progress)

- Specializing in Distributed Systems, Operating Systems, Computer Networks, and AI Systems Design. Active contributor to student engineering and hackathon teams.

Central Board of Secondary Education (CBSE) — Higher Secondary Certificate (12th Grade) *2023 - 2025*

Field: Science (PCM) | Score: 83.4%

- Strong foundation in advanced calculus, mechanics, electromagnetic physics, and algorithmic logic.

Central Board of Secondary Education (CBSE) — Secondary School Examination (10th Grade) *2021 - 2023*

Field: General Science & Mathematics | Score: 96.0%

- Academic excellence award in Mathematics and Science.

ACHIEVEMENTS & SECURITY RESEARCH

- **3rd Place — Deal Dome Competition:** Awarded 3rd place for designing the architectural specification and threat modeling for SecureVerse, a zero-trust enterprise security framework.
- **SummerHacks 2026 Backend Lead:** Led backend architecture for the 4-member team that designed F.R.I.D.A.Y, completing the 3-agent mediator engine in under 36 hours.

CERTIFICATIONS & ACCREDITATIONS

- **Responsible Disclosure & Security Research Recognition** — HackerOne (2025)
- **Multi-Agent Systems & LLM Architecture Foundations** — DeepLearning / AI Systems Guild (2026)